

Term Structure Scenarios and Expected Policy Shocks: a short note

Jan Alsterlind

June 1, 2026

Abstract

To be written ...

1 Introduction

Bond yields are forward looking variables and should reflect the market view of the future path of the central bank policy rate (the risk free short term interest rate). If, for some reason, a central bank pre-announce deviations from a future path that is not perceived to be consistent with the policy rule, we usually refer to this as expected policy shocks. In this paper, we present an easy way to calculate no-arbitrage scenarios for yields on longer maturity zero coupon bonds in the presence of anticipated policy shocks.

2 A framework for no-arbitrage pricing of bonds and expected policy shocks

A common method for scenario construction for long maturity bond yields starts from the assumption that the expectations hypothesis (EH) holds. Under EH, bond yields are just the forward convolution of the future policy path. Calculating the effect on a bond yield by announcing a policy path at some point in the future is just a matter of calculating an average of the policy path. However, an individual investor have to consider real world financial prices and any deviation between the so calculated bond yield and the actual bond yield is a ‘residual’ reflecting either a different view on the policy path or a duration premium. This note presents a method where the differences between expectations and duration premiums can be managed within a no-arbitrage framework when constructing scenarios for bond yields.

2.1 No-arbitrage bond yields

Within a no-arbitrage framework, the price of a continuously defined zero coupon bond with the maturity n is assumed to be an affine function of the pricing factors:

$$P_t^n = \exp(A_n + B_n x_t) \quad (1)$$

where A_n and B_n are coefficients and x_t is a vector of pricing factors. The pricing factors (x_t) are assumed to follow a standard time series dynamics:

$$x_t = \mu + \phi x_{t-1} + \sigma \epsilon_t \quad (2)$$

where μ , ϕ and $\sigma\sigma' = \Sigma$ are parameters and ϵ is a normal $\mathcal{N}(0, 1)$ disturbance term. It can be shown that the coefficients A_n and B_n are determined by the following difference equations:

$$A_n = A_{n-1} + B'_{n-1} \mu^Q + \frac{1}{2} B'_{n-1} \Sigma B_{n-1} + A_1 \quad (3)$$

$$B'_n = B'_{n-1} \phi^Q + B_1 \quad (4)$$

where $\mu^{\mathbb{Q}}$ and $\phi^{\mathbb{Q}}$ are the conditional mean and mean reversion of the pricing factors under the risk-neutral measure $\mathbb{Q}$. The A_1 and B_1 are the initial conditions for the difference equations, usually assumed to contain parameters describing the un-conditional mean and loadings for the pricing factors to short term interest rate (the risk free rate) of the term structure. In a basic term structure model, where the risk free interest rate is assumed to be the only pricing factor, A_1 , is the un-conditional mean for the risk free interest rate and B_1 is equal to one. We now turn to the question of how expected (anticipated) shocks can be managed within this framework.

2.2 No-arbitrage bond yields with expected policy shocks

When a central bank pre-announces future changes in the policy rate, it creates a deviation from the expected path of the pricing factors (in a simple term structure model, the policy rate and the pricing factor are often one and the same). Assume that the announced path $\tilde{x}_t$ differ from the expected, or model consistent, path x_t by an additive term:

$$\tilde{x}_t = x_t + u_t \quad (5)$$

where u_t is the known deviation from the ‘expected’ (‘model consistent’). This relationship holds for every point in time so $x_k = \tilde{x}_k + u_k, \forall k$. The inclusion of a new pricing factor (deviations from ‘model consistent’ expectations of the pricing factors) changes how the bond price is calculated. This method is originally used by Li and Wei (2014) [1] to describe the effect on bond prices by large scale asset purchase programs that is pre-announced to the market by the central bank. In this particular case, the bond price will be:

$$\tilde{P}_t^n = \exp(\tilde{A}_n + \tilde{B}_n \tilde{x}_t) \quad (6)$$

and the difference equations are:

$$\tilde{A}_n = \tilde{A}_{n-1} + \tilde{B}'_{n-1} \mu^{\mathbb{Q}} + \frac{1}{2} \tilde{B}'_{n-1} \Sigma \tilde{B}_{n-1} + \tilde{A}_1 + \tilde{B}'_{n-1} (u_{t+1} - \phi u_t) \quad (7)$$

$$\tilde{B}'_n = \tilde{B}'_{n-1} \phi^{\mathbb{Q}} + \tilde{B}_1 \quad (8)$$

Within this framework, policy shocks can not be indefinite and are assumed to last until T . In the absence of policy shocks $u_n = 0$ so in theory, there is a bond of maturity n where $n > T$. It then follows that $B_n = \tilde{B}_n$. By recursive substitution, the difference between the two bond prices are:

$$\tilde{A}_n = A_n + \sum_{i=1}^{\min(T,N)} \tilde{B}'_{n-i} u_{t+i} - \sum_{i=1}^{\min(T,N)} \tilde{B}'_{n-i} \phi u_{t+i-1} \quad (9)$$

$$\tilde{B}_n = B_n \quad (10)$$

hence, in terms of bond yields the difference can be expressed as:

$$\tilde{y}^n_t - y^n_t = b_n u_t + \sum_{i=1}^{\min(T,N)} \frac{n-i}{n} b_{n-i} (u_{t+i} - \phi u_{t+i-1}) \quad (11)$$

The intuition behind this formula is that the difference between the bond yields, at a given point in time, is the expected shock, weighted by b_n , at that time plus the future evolution of shocks ($u_{t+i} - \phi u_{t+i-1}$), weighted by $(\frac{n-i}{n} b_{n-i})$. In the case where $\phi \approx 1$, the $(u_{t+i} - \phi u_{t+i-1})$ collapses to the future change of policy shocks, Δu_{t+i} . When future policy shocks become known, bond yields jump as this information is revealed, and policy shocks can only affect bond yields if they are credible. The formula in equation (11) encapsulates the expectations hypothesis as a special case, when the duration premium is absent. Hence, scenarios for the long-term maturity bond yield can be constructed under both the risk-neutral measure $\mathbb{Q}$ but also the physical measure of the real world $\mathbb{P}$.

2.3 A simple example of expected policy shocks

As a simple example of this framework, let us consider a central bank that pre-announce that the policy rate will be different from the expected path in two years time, and thus deviate from the expected path over an extended period of eight years. After that, the policy rate will gradually return to the expected path. The effects of this pre-announcement on bond yields with different maturities are presented in Figure 1.¹

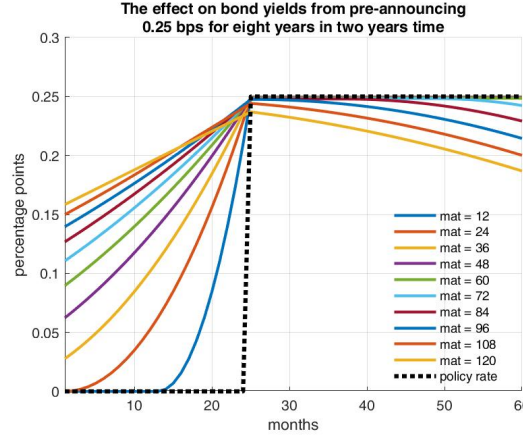


Figure 1: Expected interest rate hike in two years time

In Figure 1, longer maturity bond yields increase with announcement and continue to gradually increase until the time of the implementation of the policy shock. The one year maturity bond yield starts to increase only after 12 months after the announcement and is roughly 25 basis points higher in the second year when the shock is implemented. Longer maturity bond yields gradually begin to decline after the shock is implemented in the second year.

3 A multi-factor term structure model and expected policy shocks

In the previous section, we showed a simple example of how to construct a scenario for long-maturity bonds yields when the central bank pre-announce policy shocks in the future. In this section we describe how expected policy shocks can be implemented in multi factor no-arbitrage term structure model. In order to do this, we estimate a term structure model with four pricing factors on monthly data for Swedish (switch to US) treasuries, using a sample period from January 1994 to June 2024. The pricing factors are assumed to be observable and are rotated so one factor is closely related to the one month treasury yield. This factor represents the US policy rate when constructing policy shocks. Using observable pricing factors, the model can be estimated with linear methods (ordinary least squares) but we prefer to use a-priori information for the long run level of the short rate so a Bayesian estimation technique is used instead. The long run level of the short rate is assumed to be 2.5 percent.

3.1 Expected policy shocks and effects to bond yields

We start by showing the effect of announcing a higher path of one percent over three years for the short term interest rate at the beginning of the scenario period. After three years, it is assumed that the announced deviation reverts back to the model forecast we use as a baseline to represent the market view of the policy path. In Figure 2 we see that the scenario (that is, the conditional forecast) of the risk neutral 5 year zero coupon interest rate is revised upwards compared to the baseline scenario. As the duration premium is countercyclical to policy shocks in the estimated model, the real world scenario

¹Here, we use a simple model with one factor and with $\phi = 0.975$ and $\Sigma = 0.000075$

(including the duration risk premium) of the 5 year zero coupon interest rate is revised downward compared to the baseline in Figure 2.

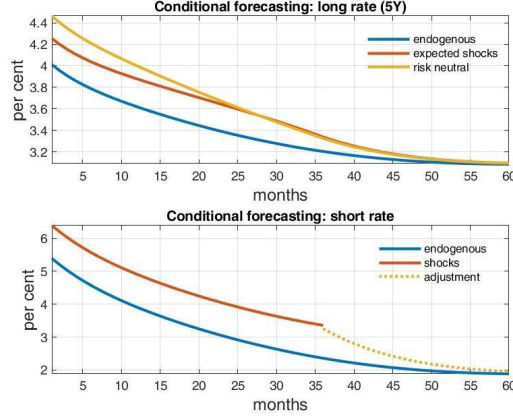


Figure 2: Expected interest rate hike of 1 percent in the beginning of the forecasting period

We also construct a scenario by assuming that the higher path is pre-announced one year in advance and that the expected deviation is to last for two years. As in the previous scenario, the path of the short rate is assumed to eventually revert back to the baseline (the unconditional forecast from the model) for the short term interest rate. In Figure 3, we see that the scenario of the 5 year zero coupon interest rate is more subdued compared to the previous scenario in Figure 2. The lower panel in Figure 3 show the path of the short term interest rate, compared to the baseline and for the first year, the scenario for the short term interest rate follows the baseline but deviates sharply in the beginning of the second year to revert back slowly to the baseline after three years.

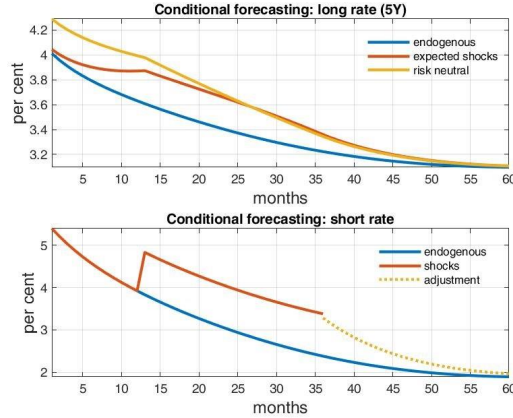


Figure 3: Expected interest rate hike of one percent in one years time is pre-announced in the beginning of the forecast period

4 Discussion

The framework shown in this note does not claim to necessarily make realistic scenarios for the cases where the central bank policy path deviates from market expectations. We use strong assumptions that the deviations announced by the central bank are completely credible and that the central bank is fully committed to the policy. However, the framework offers a simple method for scenario analysis for bond yields where the effect of expectations of the policy path and the effect of duration premiums are kept separate. The method can also be applied to all linear term structure models where policy shocks can be identified.

References

- [1] C. Li and M. Wei, “Term structure modeling with supply factors and the federal reserve’s large scale asset purchase programs,” 2014.